

ON-Delay Timer Applications

Notes from Pages 14–16

1. ON-Delay Timer with NCTO Contact (Alarm / Horn Control)

An **NCTO (Normally Closed, Timed Open)** contact remains closed when the timer coil is de-energized and opens only after a preset delay once the timer coil is energized.

Hardwired Relay Operation (Page 14)

- A momentary start pushbutton energizes the control relay.
- A sealing contact keeps the relay energized after the pushbutton is released.
- The horn turns ON immediately through a normally closed contact.
- The ON-delay timer is energized simultaneously.
- After the preset time expires, the NCTO timed contact opens.
- The horn circuit breaks and the horn turns OFF automatically.

Industrial Example: Warning horn sounds during machine startup and turns OFF automatically after a fixed delay.

2. PLC Implementation of NCTO ON-Delay Control

- Pushbutton input energizes an internal relay.
- Internal relay seals itself in the program.
- TON instruction starts timing.
- Output horn remains ON during timing.
- When the timer finishes, logic opens the horn circuit.

In PLCs, NCTO behavior is implemented using timer logic and status bits rather than physical contacts.

3. ON-Delay Timer in Automatic Sequential Control (Page 15)

Sequential control is a control strategy where multiple operations occur in a fixed order with defined conditions and time delays.

System Description

The system consists of:

- Lube-oil pump motor (M1)
- Main drive motor (M2)
- Feed motor (M3)
- Pressure switch (PS1)
- ON-delay timer (TD)

Sequence of Operation

- Start pushbutton energizes M1 (lube pump).
- M1 seals itself and continues running.
- Oil pressure builds up and closes pressure switch PS1.
- PS1 energizes M2 (main motor) and the ON-delay timer.
- Timer counts the preset delay.
- After timing completes, M3 (feed motor) is energized.

Purpose: Ensures lubrication and stabilization before applying mechanical load.

4. PLC Ladder Logic for Sequential Control (Page 16)

PLC Program Operation

- Start pushbutton energizes M1 and seals it.
- Pressure switch PS1 enables M2 and starts the TON timer.
- During timing, the feed motor remains OFF.
- When the timer done (DN) bit becomes TRUE, M3 is energized.
- A reset instruction clears all outputs and resets the timer.

Timer enable and sealing logic ensures the timer completes its delay without interruption.

5. Hardwired vs PLC-Based Sequential Control

Hardwired System	→	PLC System
Physical relays		Internal relays
Timer relay		TON instruction
Timed contact		Timer DN bit
Extensive wiring		Software-based logic

Key Exam Points

- NCTO contacts are used to turn outputs OFF after a delay
- ON-delay timers enable safe and controlled startup
- Sequential control ensures correct order of operation
- PLC implementation reduces wiring and increases flexibility
- Timer done bits trigger subsequent stages in a sequence